

- (1) Write a Matlab or Python code that uses the frequency sampling method to design a $(2k+1)$ -point discrete differentiation filter $H(F) = j2\pi F$ when $-0.5 < F < 0.5$ (k is an input parameter and can be any integer).

$$H_A(F) = j2\pi F \quad \text{for } 0 \leq F \leq 0.5$$

$$H_A(F) = j2\pi(F-1) \quad \text{for } 0.5 \leq F < 1$$

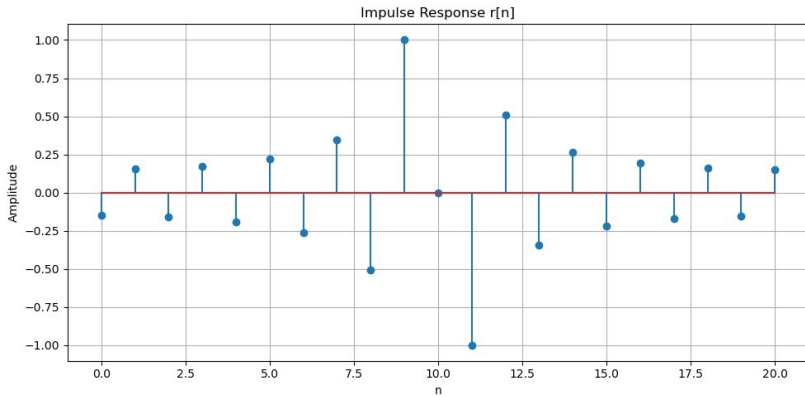
(25 scores)

The transition band is assigned to reduce the error (unnecessary to optimize).

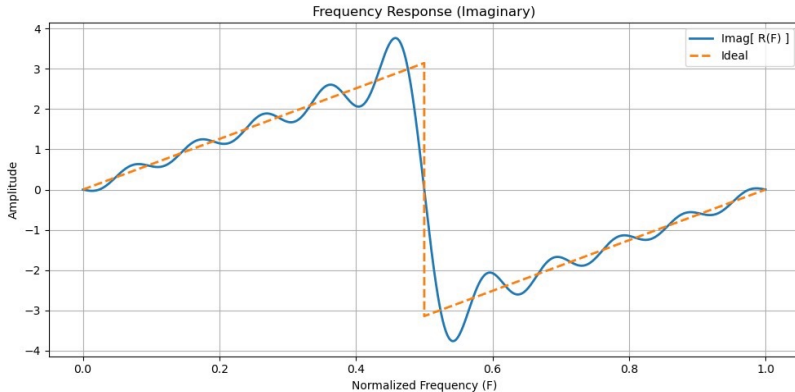
- (i) The impulse response and (ii) the imaginary part of the frequency response (DTFT of $r[n]$, see pages 112 and 113) of the designed filter should be shown in the homework. The code should be handed out by [NTU Cool](#).

$$e.x. \quad k = 10, \quad N = 21$$

(i)



(ii)



↑ — — — 訊號變化量大。

- (2) (a) What are the two main advantages of the minimum phase filter compared to other IIR filters?
 (b) What is the advantage of the Hilbert transform compared to the difference for edge detection?
 (c) What is the advantage of the Wiener filter compared to pass-stop band filters for noise removal?
 (d) What are the two advantages of the cepstrum compared to the equalizer for multipath problems?

- (a) 1. It makes both the forward and the inverse transforms stable.
 2. It makes the impulse response concentrated around 0.

(b) $y[n] = \sum_r x[n-r] h[r]$

if $h[n] = -h[n]$ (odd 對稱.)

$y[n] = h[1] (x[n-1] - x[n+1]) + h[2] (x[n-2] - x[n+2]) + \dots$

考慮的點的差異比 difference 的多。在有雜訊、無雜訊的狀況下。
 在邊緣的地方都會有比較大的強中。可以避免雜訊的影響。

- (c) 在 passband 的部分，響應會稍微小於 1。有些訊號不會完全通過。
 在 stopband 的部分，響應會稍微大於 0。有些訊號不會完全去除。

(d) equalizer 在 multipath 會有兩個問題。cepstrum 則不會有

① $H(z)$ 可能 unstable.

② $H(z)$ is usually a dynamic response.

(3) Why it is improper to use $\text{IFFT}(\text{FFT}(x[n])H(F))$ for filter design in practice? (5 scores)

$\text{IFFT}(\text{FFT}(X(n))H(F))$ 不適合的原因：

- 會產生無限長的 filter 脈衝響應。
- 在某些取樣點會產生 Gibbs phenomenon.

(4)(a) What is the role of the weight function in FIR filter design? (b) Can the techniques of the weight function be applied in the FIR filter designed by (i) the MSE method and (ii) the frequency sampling method? (10 scores)

(a) weight function 可以決定 transition band 誤差大小。

(b) (i) MSE method 可以用 weight function,

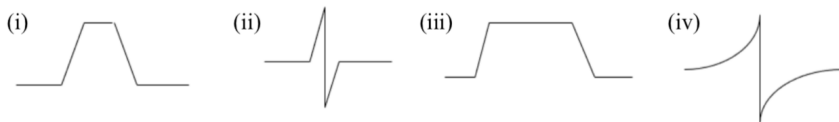
$$MSE = \int_{-\frac{1}{2}}^{\frac{1}{2}} W(F) |H(F) - H_d(F)|^2 dF$$

$$\sum_{r=0}^k s(r) \underbrace{\int_{-\frac{1}{2}}^{\frac{1}{2}} W(F) \cos(2\pi r F) \cos(2\pi r F) dF}_b = \underbrace{\int_{-\frac{1}{2}}^{\frac{1}{2}} W(F) H_d(F) \cos(2\pi r F) dF}_c, \quad \begin{matrix} r = 0 \sim k \\ n = 0 \sim k \end{matrix}$$

再用 matrix operation 表示 $S = B^T C$

(ii) frequency sampling method 不需要用 weight function, 直接在某些頻率上取樣。
所有取樣點的權重一致。

(5) The following figures are the impulse responses of some filters. Which one is a suitable **edge detector** when we want to extract (a) small scaled features? (b) large scaled features? Also illustrate the reasons. (10 scores)



(a) (ii) 較窄的脈衝響應易受雜訊影響, 無法提取大範圍特徵。
適合 small scaled edge detect.

(b) (iii) 較寬的脈衝響應有平滑性, 能更好地響應 large scaled
抑制高頻雜訊。

- (6) If the z-transform of $h[n]$ is $H(z) = \frac{2z^3 - 2z^2 - 3z - 2}{z^2 - 0.7z + 0.1}$
- (a) Convert the IIR filter into the minimum phase filter.
- (b) Determine the cepstrum of $h[n]$.

(20 scores)

$$\begin{aligned} (a) \quad H(z) &= \frac{2z^3 - 2z^2 - 3z - 2}{z^2 - 0.7z + 0.1} \quad z \times \frac{z - (\bar{z}_z^{-1})}{z - z_z} \\ &= \frac{z(\bar{z} - z)(z + (0.5 + 0.5j))(z + (0.5 - 0.5j))}{(z - 0.5)(z - 0.2)} \times (z) \frac{z - \frac{1}{z}}{z - z} \\ &= \frac{4(z + (0.5 + 0.5j))(z + (0.5 - 0.5j))}{(z - 0.2)} \end{aligned}$$

$$\begin{aligned} (b) \quad (z - 0.5)(z - 0.2) &= (1 - 0.5z^{-1})(1 - 0.2z^{-1})z^2 \\ H(z) &= \frac{4z^{-1}(\frac{1}{2}z - 1)(1 + (0.5 + 0.5j)z^{-1})(1 + (0.5 - 0.5j)z^{-1})}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})} \cdot z \end{aligned}$$

$$h[n] = \begin{cases} \log 4 & , n=0 \\ -\frac{(0.5 + 0.5j)^n + (0.5 - 0.5j)^n + \frac{0.5^n + 0.2^n}{n}}{n} & , n > 0 \\ \frac{0.5^{-n}}{n} & , n < 0 \end{cases}$$

- (7) Suppose that the cepstrum of a signal $x[n]$ is $\hat{x}[2] = 0.8$, $\hat{x}[n] = 0$ otherwise.

Determine $x[n]$ using the Z transform and $\exp(\cdot)$.

(10 scores)

$$\hat{x}[n] \longrightarrow Z[\cdot] \xrightarrow{\hat{x}(z)} \exp(\cdot) \xrightarrow{x(z)} Z^{-1}[\cdot] \longrightarrow x[n]$$

$$\hat{x}(z) = 0.8z^{-2}, \quad x(z) = e^{\hat{x}(z)} = e^{0.8z^{-2}}$$

$$\begin{aligned} x(z) &= e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \Rightarrow e^{0.8z^{-2}} = \sum_{k=0}^{\infty} \frac{(0.8z^{-2})^k}{k!} \\ &= 1 + 0.8z^{-2} + \frac{(0.8z^{-2})^2}{2!} + \frac{(0.8z^{-2})^3}{3!} + \dots \end{aligned}$$

$$x[n] = \delta[n] + 0.8\delta[n-2] + \frac{0.8^2}{2!}\delta[n-4] + \frac{0.8^3}{3!}\delta[n-6] + \dots$$